Cutting COD returns without cutting growth

To: Founder, Oytra · **From:** Kondepati Ratan · **Date:** 30 Oct 2026 **Re:** what the order data says about RTO, and what I recommend we do

The short version. Cancelling risky COD orders costs us more than it saves. A delivered ₹649 order earns ₹402 of margin, while an RTO costs ₹169. So cancelling only pays when an order is **more than 70% likely to fail**, or **81% for a first-time buyer** once we count the ₹300 we paid to acquire them. Almost no orders are that clear-cut. The money is in *confirming* and *nudging to prepaid*, not cancelling. On September's orders my rules would have added about **₹16,000 a month** at our volume (1,650 COD orders). The "cancel anything risky" rule would have *lost* about **₹97,000**.

1. What surprised me in the data (1,987 past COD orders, Apr–Sep 2026)

1. **We can't judge pincodes, but we can judge cities.** The 1,987 orders are spread over 1,066 pincodes, and none has more than 8 orders. "Bad pincode history" is mostly noise. At city level the gap is real: Gorakhpur 62%, Bareilly 61% and Darbhanga 59% RTO against 18% in Mumbai and Pune. Bihar as a whole is at 54%.
2. **1 in 11 orders has an address no courier can use** (".", "home", "call me", or just the city name). These fail 62% of the time. **But 38% of them were still delivered**, so cancelling them all would have turned away 70 real customers. Asking for the address on WhatsApp is the better move.
3. **Customer history is the strongest signal we have.** Repeat buyers who have never refused fail only 14% of the time, against 30% for first-timers. Buyers who refused 2+ times before fail 77% of the time. 75% of COD orders come from first-timers, which is exactly why we can't simply cancel the unknowns.
4. **Our most expensive customers are our riskiest COD buyers.** First-time buyers from Meta ads refuse 34% of the time, against 22% for first-time buyers who come direct. Marketing is right that these buyers matter, and they are also where prepaid nudges pay off most.
5. **A fake name on its own means little.** The 7 orders with a name like "asdf" but a real address were all delivered. A fake name *plus* a junk address failed 73% of the time. Only that combination is held.
6. *Data note:* the export had 30 duplicate orders, 5 spellings of "RTO", 9-digit phone numbers and ₹0 orders. I cleaned these before any analysis. Also, the average COD order here is ₹920, not the ₹649 company average. COD orders may simply be larger, or the sample is skewed; worth checking.

2. Recommended rules, and what they'd have done in September

Every COD order gets an RTO risk score from a simple model trained on April–August, which ranks September's orders well (AUC 0.74). Then:

Bucket	Rule	Sept orders	Real RTO rate
Ship as-is	risk < 25%	188 (57%)	14%
Confirm on WhatsApp	25–40%, or junk address (ask for full address)	73 (22%)	33%
Nudge to prepaid (₹40 off)	40%+, or 2+ past refusals; COD still allowed	63 (19%)	49%
Hold / cancel	fake name + junk address, or risk ≥ 75% (85% for first-timers)	5 (2%)	100%

We touch 43% of orders, and that group holds **70% of all RTOs**. Only 5 orders were held, and all 5 were real RTOs, so **zero genuine buyers were cancelled**. Each decision comes with a one-line reason, e.g. *"66% risk: deliveries to Gorakhpur are often refused and came from a Meta ad. Offer a small discount to pay online..."*. Every threshold above can be changed on the app's Rules page without touching code.

3. Operations vs Marketing, in rupees (September orders, real outcomes)

Policy	Orders cancelled	Genuine buyers lost	Net vs today
Ops: cancel if risk ≥ 40%	68	32	**−₹19,383**
Ops: stop COD in the 5 worst cities	37	17	**−₹9,557**
Recommended rules	5	0	**+₹3,264**

This is a 329-order sample, about one-fifth of a real month. **Ops is right that RTO is expensive** (₹78,000 a month at our volume). But blanket cancellation throws away margin worth more than the returns it prevents. That stays true even if we ignore Marketing's ₹300 acquisition cost completely: "cancel ≥ 40%" is still −₹10,983. **Marketing is right that first orders matter**, so first-timers are never cancelled on risk alone (85% bar). Instead they get confirmation and a prepaid offer, which never takes COD away. Stopping COD in bad cities stays negative even if half those buyers switched to prepaid (−₹3,084).

4. What could go wrong, and how we'd notice

- **My WhatsApp assumptions could be off.** I assumed 40% of would-be RTOs are caught, 3% of genuine buyers drop off and 20% take the prepaid offer. In a pessimistic case (25% / 6% / 10%) the system roughly breaks even (−₹312 in the sample). To find out, keep a **10% random control group** that ships without confirmation, and measure the real numbers in two weeks.
- **The model gets stale or meets new cities.** Unknown cities fall back to the state average. The dashboard shows each bucket's real RTO rate; if "Ship as-is" rises above ~18%, retrain.
- **Customers tap "Yes" and still refuse.** We'll track RTO among confirmed orders. If it stays near 33%, confirmation isn't filtering anyone.
- **The LLM misreads an address.** An LLM label can only push an order into *confirmation*, never into cancellation, and 20 labels a week get a spot-check.
- **Discount habit.** We'll watch whether repeat customers start waiting for the ₹40 offer.

5. With two more weeks, I would build

1. **Live plumbing:** a Shopify order webhook, the WhatsApp Business API (Meta Cloud API or Interakt), a payment-link webhook, and the courier's RTO feed so outcomes flow back automatically.
2. **The control-group test** above, plus an A/B test of ₹30 / ₹40 / ₹60 discounts, to replace my assumptions with our own numbers.
3. **Address repair before shipping:** an LLM rewrites Hinglish landmark addresses into a clean format, a geocoder checks them, and the customer approves the result on WhatsApp.
4. **Courier choice by city:** Delhivery's RTO rate is 26% against 31% for Xpressbees. Routing risky cities to the best courier there is a cheap win to test.
5. **Weekly automatic retraining** with a short "what changed" note to ops.